Aditya Prasanna

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Objective

I am a PhD student in Simon Fraser University, I am currently working as a Research assistant in Systems, Networking and Architecture Research Lab, SFU under the supervision of Prof. Arrvindh Shriraman. My area of research involves studying domain specific accelerators and developing intelligent compiler primitives to aid in accelerating complex memory fetch algorithms such as sparse data structure (For ex: B+Trees, Hash Tables etc.) walks.

Education

PES University – Bangalore, India 2017-2021
B.Tech Computer Science, GPA 8.09/10.00
Simon Fraser University – Burnaby, Canada 2021-2024
M.Sc. (Thesis) Computing Science, GPA 3.61/4.33
Simon Fraser University – Burnaby, Canada 2024-Present
PhD Computing Science, GPA 3.84/4.33

Skills

- **Programming Languages**: C++, C, Python, Go, R
- **Tools and Environments**: MATLAB, Tableau, Gem5, Gem5-SALAM, Champsim
- **Web Technologies**: Angular JS, Node JS
- **Compiler Technologies**: LLVM, MLIR, TVM
- **Parallel Frameworks**: CUDA, OpenMP, OpenCL
- **Databases**: MySQL, Postgres, MongoDB

Experience

Simon Fraser University

Compiler Research Engineer Intern – *Heterogeneous Compiler Lab, Huawei*
Markham, ON Canada Jan 2024 – Jun 2024

- Part of the GPU compiler team. version control and debugging of compiler builds.
- building a thread-pool manager to optimize and synchronize multithreaded compilation.

Research Assistant – *Systems, Networking and Architecture Research (SYNAR) Lab, SFU*
Burnaby, Canada Sep 2021 – Present

- Designing abstractions that expose structural locality in index traversals to guide cache- and traversal-level optimizations.
- Intelligent cache system design for Domain Specific Architectures, Simulation Modelling using gem5-SALAM (Frontend (HLS): LLVM IR, Backend (Simulation): Gem5).
- Simulation of domain specific architectures to study data reuse patterns in sparse data structure walks, analyzing cache behavior and performance due to indirect memory accesses and pointer chasing algorithms.
- Courses Taken: Computer Architecture, Parallel and distributed Computing, Database Systems, Design and Analysis of Algorithms, Advanced Topics in Computer Architecture, Machine Learning, NLP.

Teaching Assistant – *Simon Fraser University*
Burnaby, Canada Sep 2021 – Present

- CMPT 295: Introduction to Computer Systems
- In charge of handling Labs on Basic Programming C, Memory hierarchy and organization, Cache Protocols, RISC-V Assembly.
- CMPT750: Computer Architecture
  Lab instructor for gem5-SALAM

Cisco Systems, Inc. – *Technical Undergraduate Intern*
Bangalore, India Jan 2021 – July 2021

- Worked on creating MEAN applications for various scenarios. For Ex: Login Pages with role-based views, dashboards and various other feature-based views.
- Worked on integrating AWS S3 and AWS CloudWatch with NodeJS backend for logging and search-based log views.

Indian Institute of Science (IISc) – *Research Intern*
Bangalore, India Jun 2020 – Dec 2020

- Developed a cache coherency within the proprietary hardware simulator using Go.
- Designed and developed a self-defined language using the MLIR framework.
- Integrated the OpenCL pass within the TVM Deep Learning workflow for the REDEFINE processor.

Projects

Simon Fraser University

Loom: Capturing Locality in Uncertain Data-structures in Remote Memory

- Developing probabilistic structural representations that predict traversal reuse in pointer-centric data structures, enabling transformation-based optimizations that go beyond traditional hotness, spatial, and temporal locality.

CATALYST: Virtually Disaggregating Data-Structures in Far and Near Memory

- Designed a lightweight C++ runtime that reduces pointer-chasing overhead in large index structures by enabling traversal short-circuiting through range-tagged cursors.
- ***Impact.*** Cutting remote memory traffic by up to ~80% across far-memory(RDMA) and NUMA workloads.

Range-Blocks: A Synchronization Facility for Data-Structures in Domain-Specific Architectures

- Benchmarked a series of workloads on mutating graphs and indexed structures
- Setup and experimentation of workloads on various performance metrics in Gem5
- ***Submitted and Accepted @ 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25)***

METAL: Caching Multi-level Indexes in Domain Specific Architectures

- Developed patterns to identify and translate index reuse to caching pragmas
- Developed a novel Index Cache structure to support index-reuse patterns in Hierarchical indexed data structures such as B+Trees.
- ***Submitted and Accepted @ 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'24)***

Publications

1. Kumar, Anagha Molakalmur Anil, Aditya Prasanna, Jonathan Balkind, and Arrvindh Shriraman. "METAL: Caching Multi-level Indexes in Domain-Specific Architectures." In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, pp. 715-729. 2024. [LINK]

2. Kumar, Anagha Molakalmur Anil, Aditya Prasanna, and Arrvindh Shriraman. "RANGE-BLOCKS: A Synchronization Facility for Domain-Specific Architectures." *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 1*. 2025. [LINK]